

File Descriptors:

```
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <sys/types.h>

int main()
{
    int fd[2];
    pid_t pid;
    char message[] = "Hello from Parent!";
    char buffer[100];

    // Create pipe
    pipe(fd);

    // Create child process
    pid = fork();

    if (pid > 0)
    {
        // Parent process
        close(fd[0]); // Close reading end
        write(fd[1], message, strlen(message) + 1);
        close(fd[1]); // Close writing end
    }
    else if (pid == 0)
    {
        // Child process
        close(fd[1]); // Close writing end
        read(fd[0], buffer, sizeof(buffer));
        printf("Child received: %s\n", buffer);
        close(fd[0]); // Close reading end
    }

    return 0;
}
```

```
#include <stdio.h>
#include <sys/types.h>
#include <sys/stat.h>

int main()
{
    if (mkfifo("myfifo", 0666) == -1)
    {
        perror("mkfifo");
        return 1;
    }

    printf("FIFO created\n");

    return 0;
}
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("myfifo", O_RDONLY);

    read(fd, buffer, sizeof(buffer));

    printf("Received: %s\n", buffer);

    close(fd);

    return 0;
}
```

```
bhanu@DESKTOP-DE32HQ8:~$ mkdir -p ~/pipes
cd ~/pipes
bhanu@DESKTOP-DE32HQ8:~/pipes$ pwd
/home/bhanu/pipes
bhanu@DESKTOP-DE32HQ8:~/pipes$ nano anonymous_pipe.c
bhanu@DESKTOP-DE32HQ8:~/pipes$ gcc anonymous_pipe.c -o anonymous_pipe
bhanu@DESKTOP-DE32HQ8:~/pipes$ ./anonymous_pipe
Child received: Hello from Parent!
bhanu@DESKTOP-DE32HQ8:~/pipes$ ./anonymous_pipe
Child received: Hello from Parent!
bhanu@DESKTOP-DE32HQ8:~/pipes$ nano fifo_create.c
bhanu@DESKTOP-DE32HQ8:~/pipes$ gcc fifo_create.c -o fifo_create
bhanu@DESKTOP-DE32HQ8:~/pipes$ ./fifo_create
mkfifo: File exists
bhanu@DESKTOP-DE32HQ8:~/pipes$ ls -l
total 40
-rwxr-xr-x 1 bhanu bhanu 16272 Aug 29 05:31 anonymous_pipe
-rw-r--r-- 1 bhanu bhanu 722 Aug 29 05:31 anonymous_pipe.c
-rwxr-xr-x 1 bhanu bhanu 16056 Aug 29 05:32 fifo_create
-rw-r--r-- 1 bhanu bhanu 220 Aug 29 05:32 fifo_create.c
prw-r--r-- 1 bhanu bhanu 0 Aug 29 04:10 myfifo
bhanu@DESKTOP-DE32HQ8:~/pipes$ p
p: command not found
bhanu@DESKTOP-DE32HQ8:~/pipes$ nano fifo_read.c
bhanu@DESKTOP-DE32HQ8:~/pipes$ gcc fifo_read.c -o fifo_read
bhanu@DESKTOP-DE32HQ8:~/pipes$ nano fifo_write.c
bhanu@DESKTOP-DE32HQ8:~/pipes$ gcc fifo_write.c -o fifo_write
bhanu@DESKTOP-DE32HQ8:~/pipes$ ./fifo_read

Received: Hello from Writer
bhanu@DESKTOP-DE32HQ8:~/pipes$
bhanu@DESKTOP-DE32HQ8:~/pipes$ |
bhanu@DESKTOP-DE32HQ8:~$ cd ~/pipes
bhanu@DESKTOP-DE32HQ8:~/pipes$ cd ~/pipes
bhanu@DESKTOP-DE32HQ8:~/pipes$ ./fifo_write
Message sent
bhanu@DESKTOP-DE32HQ8:~/pipes$ |
```